QuickPhone — Level 0 DFD (physical)

Human–machine boundary: QuickPhone web app and its database

External entities

- Front Desk Staff
- Repair Technician
- Store Manager
- Customer
- Store Owner

Processes

3.0 Record Diagnosis — *Web app · Technician · Front Desk*

1.0 Schedule Appointment — *Web app booking screen · Front Desk · counter PC*

2.0 Register Repair Job — *Web app + label printer · Front Desk · counter PC*

5.0 Send Completion Notice — *Web app → store email account (SMTP)*

6.0 Record Device Release — *Web app; card machine not linked*

8.0 Report Repair Status — *Web app dashboard · Owner · Front Desk*

4.0 Record Repair Work — *Web app · Technician · bench PC*

9.0 Maintain Shop Records — *Web app admin screens, CSV import, nightly backup*

7.0 Process Purchase Orders — *Web app; orders placed by phone or supplier website*

Data stores

- **D4 Quotes** — quote
- **D3 Repair Jobs** — repair_job, job_status, job_status_history, job_note, job_part
- **D7 Parts Inventory** — part, store_part_stock, inventory_txn
- **D2 Appointments** — appointment
- **D10 Staff & Stores** — employee, role, store
- **D1 Customers & Devices** — customer, device
- **D9 Suppliers** — supplier
- **D6 Notifications** — notification
- **D5 Payments** — payment
- **D8 Purchase Orders** — purchase_order, po_line, po_job_link
- **D11 Backup Copies** — backup_log + backup files

Flows

Quote decision entry; Awaiting Approval queue; My job queue & stock on hand; Job click, diagnosis & quote forms; quote row & decision; Pending quote rows; repair_job rows (mine); job_note, status & job_part rows; store_part_stock & standard_cost; quote.total_amount; repair_job (In Repair); job_part, notes & status rows; Repair job screen & Awaiting Parts alert; Job click, parts-used & completion forms; Booking form entry; Free-slot screen & confirmation; repair_job + history (Checked In); appointment rows; appointment row; customer row; Active technicians; quantity_on_hand; Stock −qty & inventory_txn (Used); part & stock rows; part, stock & inventory_txn; Nightly; Diagnosing or In Repair status; customer rows; appointment row; appt_status update; Technicians at this store; employee & store rows; employee & store rows saved; Phone search & intake forms; Search results, printed ticket & tag label; customer rows final cost; customer & device rows; All tables, nightly (D1–D10); Jobs awaiting parts; Admin forms & CSV import file; Save confirmation & import report; Ready for Pickup jobs; repair_job & history rows; Stock at or below reorder point; supplier rows; supplier row; Store hours & sending address; Final cost entry & Send click; customer.email; store_part_stock; Stock +qty & inventory_txn (Received); customer rows; Ready for Pickup or Returned Unrepaired job; Ready-to-notify list & email preview; notification rows; notification row; store rows; supplier rows; picked_up_at, released_by & Closed status; Backup file & log; payment row; payment row (exists?); Customer name; Ticket lookup, ID tick, payment & signature; PO form, placed & received entries; PO & line rows; PO line & link rows; Pickup screen & release receipt; notification rows; Low-stock, PO status & released jobs; Backup status; Status search; Job status screen; Completion email (SMTP); Two-store dashboard (screen); Approve / Reject + reason; PO approval queue

Physical DFD symbols (Gane and Sarson)

- External entity
- Automated process *(bottom band = how / who)*
- Manual process
- Human–machine boundary
- Data store table names
- Data flow (one way)
- Timer trigger (control)
- Converging flow *(every symbol drawn once)*

Physical view adds implementation bands, the human–machine boundary, table names in each store, the nightly backup inside 9.0 and D11 Backup Copies.
Each store holds the ERD entities listed in it; every ERD entity sits in exactly one store (see the store–entity matrix). 8.0 is a primitive covered by its process description.